

How Fast Does the Infinite Sinc Product Decay?

A gentle guide to the answer, for readers who know limits, integrals, and a little Fourier analysis

Companion to the *Comparative Audit* in the same directory
(which contains the precise statements, proofs, and literature forensics)

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TL;DR

The function. $f(x) = \prod_{n=0}^{\infty} \operatorname{sinc}(\pi x/2^n)$, where $\operatorname{sinc} z = \frac{\sin z}{z}$. It is the Fourier transform of the “smoothest possible bump”: an infinitely differentiable function that is zero outside $[-1, 1]$. It equals 1 at $x = 0$, vanishes at every positive integer, and in between it wiggles with rapidly shrinking humps.

How fast it decays. Faster than any power of x , slower than any exponential. The precise shape is

$$|f(x)| \approx x^{-\kappa} \cdot \exp\left(-\frac{(\ln x)^2}{2 \ln 2}\right),$$

where the exponential factor is universal, but—and this is the punchline—the power κ depends on what you mean by “how big is f near x ”:

tallest humps	$\kappa_{\infty} = 2.3590 \dots$
mean square (energy)	$\kappa_2 = \log_2(2\pi) = 2.6515 \dots$
plain average of $ f $	$\kappa_1 = 2.7481 \dots$
typical point	$\kappa_0 = \frac{3}{2} + \log_2 \pi = 3.1515 \dots$

There is no single “decay rate.” Asking for one is like asking for “the income of a country”: the mean, the median, the richest person, and the root-mean-square are genuinely different numbers.

The answer to the Stack Exchange question (which asks for g such that the running average of $|f|/g$ tends to 1): take

$$g(x) = 0.09127 \cdot x^{-2.7481} \exp\left(-\frac{(\ln x)^2}{2 \ln 2}\right).$$

Then the running average of $|f|/g$ stays between 0.92 and 1.12 forever (so the “bounded” version of the question is solved), and it converges to exactly 1 if you stop the average at powers of two, or if you average “per octave” ($\frac{1}{\ln t} \int_t^{2t} \frac{|f|}{g} \frac{dx}{x}$). But for arbitrary stopping points the limit provably does *not* exist—for this g or for any other formula-like g . The average forever oscillates between about 0.92 and 1.11 depending on where between two powers of two you stop. That obstruction is a real feature of the function, not a failure of cleverness.

A bonus discovery. Cross-checking two of the AI-written solution drafts against each other exposed a genuine error in a published 2017 paper (Annales de l’Institut Fourier): a variance was computed without the factor $\frac{1}{2}$ from $\int_0^1 \cos^2(2\pi kt) \, dt = \frac{1}{2}$, making a headline constant $\sqrt{2}$ times too large (π where it should be $\pi/\sqrt{2}$). The five-line correct computation is in [section 8](#).

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1 The function, and why anyone cares

Start with the most famous function of signal processing,

$$\operatorname{sinc} z = \frac{\sin z}{z}, \quad \operatorname{sinc} 0 = 1.$$

Now multiply infinitely many copies of it, each one stretched twice as much as the last:

$$f(x) = \operatorname{sinc}(\pi x) \cdot \operatorname{sinc}\left(\frac{\pi x}{2}\right) \cdot \operatorname{sinc}\left(\frac{\pi x}{4}\right) \cdot \operatorname{sinc}\left(\frac{\pi x}{8}\right) \cdots$$

The product converges nicely: for large n the factor $\operatorname{sinc}(\pi x/2^n)$ is extremely close to 1 (since $\operatorname{sinc} z \approx 1 - z^2/6$ near 0), so past some point the factors stop mattering.

Why this particular product? Because sinc is the Fourier transform of a rectangular pulse, and multiplying Fourier transforms means *convolving* the originals. So f is the Fourier transform of the infinite convolution of rectangles of widths $1, \frac{1}{2}, \frac{1}{4}, \dots$ — a famous function called $\operatorname{up}(t)$ (Rvachev’s *up-function*, a close relative of the Fabius function): a perfectly smooth bump, infinitely differentiable, exactly zero outside $[-1, 1]$. Smooth compactly supported functions have rapidly decaying Fourier transforms, so f must die faster than any power of x . The question — asked on Mathematics Stack Exchange in 2018 and the subject of this directory — is: *exactly how fast?*

Three visible features of f , all easy to believe and all proved in the audit:

- $f(m) = 0$ for every nonzero integer m (already the first factor $\operatorname{sinc}(\pi x)$ vanishes there);
- between consecutive integers, $|f|$ makes exactly one hump;
- the *sign* of f on the interval $(m, m+1)$ is $(-1)^{(\text{number of 1's in the binary digits of } m)}$ — the Thue–Morse pattern:

m	0	1	2	3	4	5	6	7
binary	0	1	10	11	100	101	110	111
sign of f on $(m, m+1)$	+	−	−	+	−	+	+	−

Binary digits will keep showing up. That is not a coincidence: the factor scales are powers of 2, so everything about this function is secretly about binary expansions.

2 How small is it, really?

Some honest numbers, computed directly from the product:

$$|f(10.5)| \approx 10^{-5.0}, \quad |f(100.5)| \approx 10^{-13.0}, \quad |f(1000.5)| \approx 10^{-26.4}.$$

Each factor-of-ten step in x roughly *doubles* the number of zeros after the decimal point, rather than adding a fixed number. That is the signature of a law like $\exp(-c(\ln x)^2)$: the exponent is *quadratic* in $\ln x$. A plain power law x^{-p} would add the same p digits per decade; an exponential e^{-cx} would multiply the digits by ten per decade. This function sits exactly in between, and the questioner's original guess, "something close to $\exp(-\ln^2 x)$," was pointing at the right family. The correct constant turns out to be $\frac{1}{2 \ln 2} = 0.7213 \dots$ in place of 1, and on top of the $\exp(-(\ln x)^2/(2 \ln 2))$ factor there is a power of x that carries all the interesting mathematics. Let us see where each part comes from.

3 Where the $\exp(-(\ln x)^2/(2 \ln 2))$ comes from: just count

Fix a large x , say $x = 1000$, and look at the factors $\text{sinc}(\pi x/2^n)$ one at a time.

- If 2^n is *much bigger* than x , the argument $\pi x/2^n$ is tiny and the factor is essentially 1. Harmless.
- If 2^n is *smaller* than x (for $x = 1000$ that means $n = 0, 1, \dots, 9$, since $2^9 = 512 < 1000 < 1024 = 2^{10}$), the argument is large, and

$$\left| \text{sinc}\left(\frac{\pi x}{2^n}\right) \right| = \frac{|\sin(\pi x/2^n)|}{\pi x/2^n} \leq \frac{2^n}{\pi x}.$$

Each of these factors is *small*, and there are about $k+1 \approx \log_2 x$ of them.

Multiply the small ones together, ignoring the sines for a moment:

$$\prod_{n=0}^k \frac{2^n}{\pi x} = \frac{2^{0+1+\dots+k}}{(\pi x)^{k+1}} = \frac{2^{k(k+1)/2}}{(\pi x)^{k+1}}, \quad k \approx \log_2 x.$$

Now take logarithms and keep the biggest term. With $k \approx \ln x / \ln 2$,

$$\ln(\text{product}) \approx \frac{k^2}{2} \ln 2 - k \ln(\pi x) \approx \frac{(\ln x)^2}{2 \ln 2} - \frac{(\ln x)^2}{\ln 2} = -\frac{(\ln x)^2}{2 \ln 2}.$$

That is the whole story of the universal factor: *about* $\log_2 x$ factors, *each of size roughly* $2^n/x$. For $x = 1000$ this crude count predicts $|f| \lesssim 10^{-21.4}$; the actual typical size 10^{-26} is smaller because we threw away the sine factors, which brings us to the interesting part.

4 The wild part: sines and binary digits

The neglected numerators are

$$|\sin(\pi x)| \cdot |\sin(\pi x/2)| \cdots |\sin(\pi x/2^k)|,$$

a product of about $\log_2 x$ numbers, each between 0 and 1. How big is such a product? Here is the key trick. Since $|\sin(\pi t)|$ only depends on t modulo 1, only the *fractional parts* of x , $x/2$, $x/4$, $\dots$ matter. And halving a number shifts its binary digits: if

$$x = 1101101.0110101 \dots \text{ (binary),}$$

then $x/2 = 110110.10110101 \dots$, $x/4 = 11011.010110101 \dots$, and so on. Each factor $|\sin(\pi x/2^n)|$ reads the binary expansion of x *through a window shifted by n places*. A product over n of such factors is therefore a walk along the binary digits of x — and binary digits of a typical number behave like coin flips. Two consequences:

- *On average, each sine factor costs a factor $\frac{1}{2}$.* The average of $\ln |\sin(\pi t)|$ over $t \in [0, 1]$ is $-\ln 2$ (a classical integral). So a typical product of k sines is about 2^{-k} , which for $x = 1000$ ($k \approx 10$) accounts for the missing 10^{-3} between the crude bound $10^{-21.4}$ and reality $\approx 10^{-26}$ — well, most of it; the rest is bookkeeping in the crude estimate itself.
- *But the fluctuations around “typical” are enormous.* The factors are like random numbers, and a product of k random factors has a spread that grows with k : the logarithm performs a random-walk-like motion of typical size $\sqrt{k}$ around its mean (with variance $\pi^2/4$ per step — this specific constant is the star of [section 8](#)). Since multiplying by $e^{\pm c\sqrt{k}}$ means gaining or losing several *orders of magnitude*, nearby values of x can give wildly different $|f(x)|$.

There is one special binary pattern worth knowing. If the fractional part of x is $\frac{1}{3} = 0.010101 \dots$ (binary), then shifting the digits alternates between $0.0101 \dots = \frac{1}{3}$ and $0.1010 \dots = \frac{2}{3}$, and every sine factor equals $\sin(\pi/3) = \frac{\sqrt{3}}{2} = 0.866$ — the largest sustainable value. No point can keep *all* its sine factors near 1: if t is close to $\frac{1}{2}$ (where $\sin(\pi t) = 1$), then $2t$ is close to 0 or 1 (where the sine *vanishes*). Staying on the $\frac{1}{3} \leftrightarrow \frac{2}{3}$ cycle is the best compromise, and this single fact (the “Gelfond bound,” proved in the audit by a two-line algebraic inequality) pins down the tallest humps of f .

5 One octave under the microscope

Everything above becomes visible if we plot, for every unit interval $[n, n+1]$ inside one “octave” $[512, 1024]$, the height E_n of the single hump of $|f|$ there.

Three lessons from [figure 1](#):

1. **The dips are at multiples of powers of two.** Why? At $x = 2^j \cdot (\text{odd})$, not just one factor but $j+1$ factors of the product vanish simultaneously: $\sin(\pi x)$, $\sin(\pi x/2)$, $\dots$, $\sin(\pi x/2^j)$ all hit an integer argument. A zero of multiplicity $j+1$ crushes its neighborhood, so the humps next to “very even” integers are exceptionally tiny. The extreme case is the hump just before $1024 = 2^{10}$: height 10^{-35} , against $10^{-20.4}$ for the best hump of the same octave.
2. **The tallest humps live near mantissa $7/6$.** $597/512 \approx 7/6$, and $\frac{7}{6} - 1 = \frac{1}{6}$ reaches the optimal cycle $\frac{1}{3}$ after one digit shift. (As x grows, the best mantissa slowly drifts toward 1; the audit computes this drift exactly, with a Lambert- W function.)
3. **“The” size of f near $x = 700$ is not one number.** Within this single octave, the tallest hump, the average hump, the root-mean-square hump, and the median hump are already visibly different scales — and the gaps between them *grow* as a power of x . That is the content of the next section.

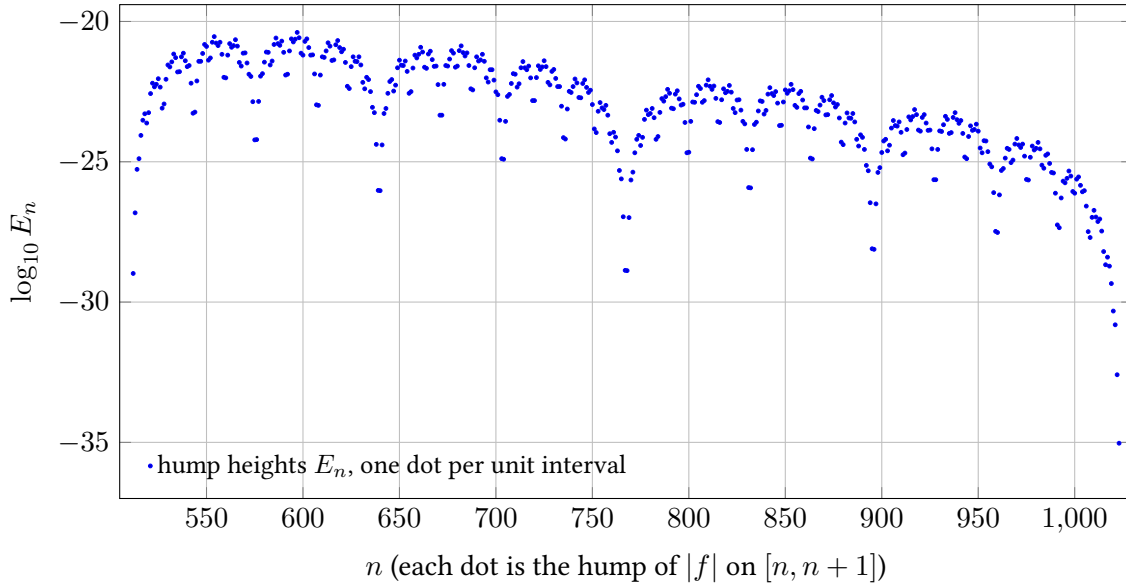


Figure 1: All 512 hump heights of $|f|$ in the octave $[512, 1024]$, on a $\log_{10}$ scale. Note the range: from $10^{-20.4}$ (near $n = 597$) down to 10^{-35} (the last hump before 1024) – *fifteen orders of magnitude inside a single octave*. The downward spikes sit at $640 = 5 \cdot 2^7$, $768 = 3 \cdot 2^8$, $896 = 7 \cdot 2^7$, ...: the more factors of 2 in n , the deeper the dip. The tallest hump sits near $512 \cdot \frac{7}{6} \approx 597$, whose fractional dynamics locks onto the optimal $\frac{1}{3} \leftrightarrow \frac{2}{3}$ binary cycle.

6 Four legitimate answers to “how fast does it decay?”

Think of the humps in an octave as incomes in a country: a few enormous ones, a broad middle class, and a long tail of tiny ones (spread over many orders of magnitude – roughly “log-normal,” like real incomes). Then:

question	analogy	answer	value of κ
biggest value near x ?	richest person	$x^{-\kappa_\infty} \exp\left(-\frac{(\ln x)^2}{2 \ln 2}\right)$	$\kappa_\infty = \frac{3}{2} + \log_2 \frac{\pi}{\sqrt{3}} = 2.3590$
root-mean-square?	RMS income	same shape	$\kappa_2 = \log_2(2\pi) = 2.6515$
plain average of $ f $?	mean income	same shape	$\kappa_1 = \frac{1}{2} + \log_2 \frac{\pi}{\varrho} = 2.7481$
typical value?	median income	same shape	$\kappa_0 = \frac{3}{2} + \log_2 \pi = 3.1515$

All four share the universal factor $\exp(-(\ln x)^2/(2 \ln 2))$ from the counting argument; they differ in the power of x , because they weigh the huge-but-rare humps differently. “Biggest” is dominated by the $\frac{1}{3}$ -cycle (that is where $\sqrt{3}$ enters). “Root-mean-square” turns out to be *exactly* computable (section 9) – that is where the neat closed form $\log_2(2\pi)$ comes from, and it is precisely the exponent that an old draft in this repository proposed, mistaking it for a pointwise decay rate. “Average” involves a constant $\varrho = 0.661322602 \dots$ with no known closed form: it is the growth rate of the integrals $\int_0^1 |\sin(\pi t) \sin(2\pi t) \cdots \sin(2^{k-1}\pi t)| dt \approx \text{const} \cdot \varrho^k$, a number studied in analytic number theory (Fouvry–Mauduit). “Typical” comes from the average-cost- $\frac{1}{2}$ -per-factor rule.

Note the moral: if you simulate the function and eyeball a slope, the answer you get depends entirely on *what you averaged before taking the slope*. Several plausible-looking published and drafted formulas for “the” decay rate of this function differ exactly because each implicitly chose a different row of this table.

7 The answer to the Stack Exchange question

The original question asks for a nice function g (positive, smooth, decreasing) such that

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx \longrightarrow 1 \quad \text{as } t \rightarrow \infty. \quad (*)$$

Averaging $|f|/g$ is asking g to be the *mean* size of $|f|$ — so, by the table, g must be the κ_1 row. The audited answer:

$$g(x) = A_1 x^{-\kappa_1} \exp\left(-\frac{(\ln x)^2}{2 \ln 2}\right), \quad \kappa_1 = 2.748070\dots, \quad A_1 = 0.0912661\dots$$

And then:

- **The bounded version of the question is fully solved.** The running average of $|f|/g$ stays between 0.92 and 1.12 for all large t . (The question offered this as an acceptable fallback — condition (3) in the original post.)
- **The limit (*) holds along** $t = 1, 2, 4, 8, \dots$: if you always stop the average at a power of two, it converges to exactly 1 (numerically it is 1.000000 from $t = 2^9$ on).
- **But for arbitrary t the limit provably fails** — for this g and for every g given by any formula of the usual kind. Instead, the average approaches a periodic-in- $\log_2 t$ profile: stopping at $t = 1.15 \cdot 2^k$ always gives about 0.92, stopping at $t = 1.43 \cdot 2^k$ always gives about 1.11.

Why can no clever g fix this? Two ingredients.

First ingredient: the last octave never fades. In a running average up to t , the chunk between $t/2$ and t occupies half the interval — always. So the average never stops caring about the most recent octave, and if the mass of $|f|$ is distributed unevenly *inside* each octave in a way that repeats from octave to octave, the average inherits that unevenness forever. (Contrast a function whose “pattern length” stays fixed while $t \rightarrow \infty$: then any one pattern eventually occupies a vanishing fraction of $[0, t]$ and averages out. Here the pattern is multiplicative — it doubles with t — so it never dilutes.)

Second ingredient: inside one octave, the mass distribution is fractal. One might hope to build the octave profile into g . The obstruction: as $k \rightarrow \infty$ the fraction of the octave-mass of $|f|$ lying in the first 100% of the octave converges to a function $F(\theta)$ that is continuous but *singular* — it rises, like the devil’s staircase, on a set that a smooth density cannot see (most of the rise happens near mantissas related to the $\frac{1}{3}$ -cycle). A smooth g can only reweight the octave by a smooth factor, and no smooth reweighting turns a staircase into a straight line. That is the rigorous content of “the histogram never flattens,” and it is proved in the audit with a pleasantly low-tech argument (if the limiting distribution had a density, iterating the digit-shift would blow the density up beyond any integrable bound).

The repair. The mismatch is between an *additive* average (dx , which overweights the last octave) and a *multiplicative* function (whose natural period is the octave). Average multiplicatively instead — give every octave equal weight:

$$\frac{1}{\ln(t/t_0)} \int_{t_0}^t \frac{|f(x)|}{g(x)} \frac{dx}{x} \longrightarrow \text{const} \quad \text{for all } t \rightarrow \infty,$$

because now the final partial octave carries weight $\sim 1/\ln t \rightarrow 0$. With the constant absorbed into g , the exact analogue of (*) holds with no restriction on t . In one sentence: *the question as posed has answer “no, but”: no for the literal limit with any gauge of the natural, shape-stabilizing kind, yes for the bounded version, yes along powers of two, and yes for every t once the average respects the function’s own dyadic geometry.* One postscript: a second audit wave later constructed a gauge that does attain

the literal limit for all t — necessarily a *shell-adaptive* analytic gauge, one that rebuilds the fractal octave profile into itself and so lives outside the natural class the impossibility theorem speaks about. The staircase never flattens; the repaired ruler is itself staircase-shaped.

8 The missing $\frac{1}{2}$: a cautionary tale

While cross-checking the four AI-written solutions, two of them disagreed about a fluctuation constant: one derived $\pi/\sqrt{2}$, the other quoted π from a 2017 paper in the *Annales de l'Institut Fourier*. Both could not be right, and the disagreement is settled by a computation short enough to do here. The quantity is the variance of

$$S_k(t) = \sum_{j=0}^{k-1} \ln |2 \sin(\pi 2^j t)|,$$

with t drawn uniformly from $[0, 1]$ (this is exactly the logarithm of the sine product from [section 6](#), with the convenient 2's making each term average to zero). Three steps:

1. *Fourier series*. Classically, $\ln |2 \sin \pi t| = - \sum_{m \geq 1} \frac{\cos(2\pi m t)}{m}$.
2. *Variance of one term*. By Parseval — remembering $\int_0^1 \cos^2(2\pi m t) dt = \frac{1}{2} -$

$$\int_0^1 \ln^2 |2 \sin \pi t| dt = \frac{1}{2} \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{12}.$$

3. *Correlations between terms*. Term j and term $j + r$ share only the frequencies where $m = 2^r m'$, giving

$$\int_0^1 \ln |2 \sin \pi t| \cdot \ln |2 \sin(\pi 2^r t)| dt = \frac{1}{2} \sum_{m' \geq 1} \frac{1}{2^r m'^2} \cdot \frac{1}{m'} \cdot m' = \frac{\pi^2}{12} \cdot \frac{1}{2^r},$$

and summing the whole covariance array,

$$\frac{\text{Var } S_k}{k} \longrightarrow \frac{\pi^2}{12} \left(1 + 2 \sum_{r \geq 1} 2^{-r} \right) = \frac{\pi^2}{12} \cdot 3 = \frac{\pi^2}{4}.$$

So the “standard deviation per factor” is $\pi/2$, and the standard theory of such sums then gives fluctuation constants proportional to $\sqrt{\pi^2/4}$ — producing the $\pi/\sqrt{2}$ of the first document. The published paper’s equation (4.5) evaluates *the same variance* but with each Parseval $\frac{1}{2}$ replaced by 1, obtaining $\pi^2/2$ and hence constants $\sqrt{2}$ too large. The slip is verified three ways in the audit: against the publisher’s PDF, against the authors’ own $\text{\LaTeX}$ source, and — most convincingly — by simulating S_{1000} on four million exactly-computed binary orbits: measured variance 2459.5, against 2464.1 predicted by $\pi^2/4$ and 4928.2 predicted by $\pi^2/2$. Verdict: the paper’s Theorems 1.1 and 1.2 hold with their constants divided by $\sqrt{2}$. (The repository now carries, next to the transcription of the original, a corrected edition of the whole paper — `AIF_2017__67_2_637_0-corrected.tex` under `docs/papers/` — whose constants match this computation.)

Two morals for the working mathematician. First, $\int \cos^2 = \frac{1}{2}$ has claimed another victim; check your Parsevals. Second, the error was found *not* by checking a document against the literature — the second document matched the literature perfectly — but by insisting that two independent derivations agree with each other, and then measuring the disputed quantity. Faithful citation is not correctness.

9 The humps, and the mean-square miracle

Two more findings from the expanded audit, in friendly form.

Hump heights are glued to areas. Each hump is a smooth arch, so its height and its area (the integral of $|f|$ over the unit interval) should be comparable — and they are, provably within a factor $C(\ln n)^3$ and in practice much tighter: across whole octaves,

$$\frac{\text{sum of hump heights}}{\text{integral of } |f|} = 1.9237 \dots,$$

constant to five digits from $x = 512$ to $x = 32768$. Consequently the *average hump height* obeys the same κ_1 law as the average of $|f|$ — the “envelope of maxima” reading of the question and the “running mean” reading collapse into one, which is reassuring. (The typical hump, by contrast, follows the κ_0 law, and the histogram of log-heights within an octave approaches a bell curve of width $\frac{\pi}{2} \sqrt{\log_2 x}$, visibly clipped on the right by the $\frac{1}{3}$ -cycle ceiling — all of this is quantified in the audit.)

Squares are where God smiles. Replace $|f|$ by f^2 (energy, in signal-processing terms) and everything becomes exact, essentially because $\sin^2$ has the clean average $\frac{1}{2}$ and because of binary uniqueness. A sample, provable in a few lines each:

$$\int_0^1 \prod_{j=0}^{k-1} \sin^2(\pi 2^j t) dt = \frac{1}{2^k} \quad \text{exactly, for every } k,$$

(write each $\sin^2$ as $\frac{1}{2}(1 - \cos)$; expanding the product gives cosines whose frequencies are sums of *distinct* powers of two, and by uniqueness of binary representation none of them can cancel to frequency zero, so only the constant term survives); and the total energy identity

$$\int_{-1}^{\infty} f(x)^2 dx = \int_{-1}^1 \text{up}(t)^2 dt = \frac{1}{2} + \sum_{m=0}^{\infty} f\left(m + \frac{1}{2}\right)^2 = 0.8088735359679 \dots,$$

which is just Parseval’s theorem for the Fourier series of the bump up on $[-1, 1]$ (its Fourier coefficients are values of f at half-integers!). The two sides were computed independently and agree to $2 \cdot 10^{-15}$ — a pleasant end-to-end check that every formula in this story is what it claims to be. The mean-square envelope even converges at a knowable geometric rate: the relevant averaging operator has second eigenvalue exactly $-\frac{1}{4}$ (eigenfunctions $\sin 2\pi x$ and $1 + 3 \cos 2\pi x$ — a two-line verification), so the RMS constant settles with error $\sim (-\frac{1}{2})^k$ per octave, alternating in sign, precisely as the measurements show.

One last warning flows from the same circle of ideas. Since $\ln |f|$ fluctuates like a random walk, it is tempting to model $|f|$ as log-normal and predict the mean and mean-square from the bell curve. This fails badly — the bell-curve model predicts $\kappa_2 \approx -0.41$ (i.e. moment *growth*) against the true 2.65 — because the extreme humps that would dominate such averages are capped by the $\frac{1}{3}$ -cycle ceiling. The correct moment-generating object is the κ_q curve itself, which is exactly what the four-row table of [section 6](#) samples.

10 What to remember

1. f decays like $x^{-\kappa} \exp(-(\ln x)^2/(2 \ln 2))$: the exponential factor comes from *counting factors*, and is universal.
2. The power κ is not unique: biggest/mean-square/mean/typical give $2.359 < 2.651 < 2.748 < 3.151$. Any claim about “the” decay rate must say which average it means.

3. The literal limit requested on Stack Exchange is unattainable by any gauge of the natural, shape-stabilizing kind (last-octave effect + fractal octave profile); the bounded version, the powers-of-two version, and the per-octave version are all solved by $g = 0.0913 x^{-2.748} \exp(-(\ln x)^2/(2 \ln 2))$, and a later shell-adaptive analytic gauge, outside that natural class, attains even the literal limit.
4. Binary digits run the show: signs are Thue–Morse, valleys sit at multiples of powers of two, peaks lock onto the binary cycle of $\frac{1}{3}$, and the constant $\varrho = 0.6613\dots$ is a genuinely new number of this dyadic world.
5. $\int_0^1 \cos^2 = \frac{1}{2}$. Even the Annales de l’Institut Fourier forgot it once.

11 Where to find the real proofs

Everything asserted here is stated precisely and proved (or explicitly labeled as imported/conjectural) in the companion document *The Decay Envelope of the Rvachev Sinc Product: A Comparative Audit* in the sibling directory `Rvachev_Up_Fourier_Decay_Comparative_Audit/`: the exact factorization and the κ -dictionary (its §3–§5), the impossibility theorem and the logarithmic-mean repair (§5), the lobe-maxima analysis (§6), the mean-square theory (§7), the audit of the four AI-written solution drafts (§8), and the correction to the published paper, with page-level references and three independent source checks (§9). The shell-adaptive gauge that attains the literal limit is constructed in the audit’s second wave (its `sec:second-wave`, with the full analysis in the sibling directory `Rvachev_Up_Fourier_Decay_Second_Wave_Audit/`). The original questions are Mathematics Stack Exchange 2745497 (decay rate) and 2742261 (extrema); local copies of the corrected 2017 paper are under `docs/papers/`.